

Problem Set 1

Data Visualisation for Social Scientists

Due: January 28, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Wednesday January 28, 2026. No late assignments will be accepted.

Roll Call Votes in the European Parliament

Data Manipulation

First, you need to download data from the first six elected European Parliaments on each MEP and how they voted in each recorded roll-call vote.

```
1 library(readr)
2 library(readxl)
3 library(dplyr)
4 library(tidyr)
5 file_path <- "mep_info_26Jul11.xls"
6 sheets <- c("EP1", "EP2", "EP3", "EP4", "EP5")
7 mep_list <- lapply(sheets, function(sheet_name) {
8   read_excel(file_path, sheet = sheet_name, col_names = TRUE)
9 })
10 names(mep_list) <- sheets
11 rcv_ep1 <- read_csv("rcv_ep1.txt")
```

I loaded both of the datasets into R

1. Load these datasets into your global environment:

- `mep_info_26Jul11.xls` (MEP characteristics, EP1–EP5)
- `rcv_ep1.txt` (EP1 roll-call votes)

2. Briefly describe (2–3 sentences each) the unit of analysis and key variables in each of these two datasets.

In the MEP dataset the unit of analysis is the MEP and the unit of analysis in the RCV dataset is a single MEP's vote in a roll call session. The key variables are: MEP's name, member state, national party, EP Group, and scores for the MEP dataset, For the RCV dataset the key variables are how the MEPs' voted and MEP name, MEP ID, member state, national party, and EP Group.

3. The `rcv_ep1` data are in a wide format, with `V1`, `V2`, ..., `Vn` as separate vote columns.

- Identify which columns are ID/metadata (`MEPID`, `MEPNAME`, `MS`, `NP`, `EPG`) and which columns are vote decisions ($V_1 \dots V_n$). Tidy the voting data such that each row/observation is a single vote for a single MEP.

```
1 colnames(rcv_ep1)[1:5] <- c("MEPID", "MEPNAME", "MS", "NP", "EPG")
2 colnames(rcv_ep1)[6:ncol(rcv_ep1)] <- paste0("V", 1:(ncol(rcv_ep1)-5))
3
4 rcv_ep1_long <- rcv_ep1 %>%
5   pivot_longer(
6     cols = starts_with("V"),
7     names_to = "VoteNumber",
8     values_to = "Vote"
9   )
10 rcv_ep1_long <- rcv_ep1_long %>%
11   mutate(VoteLabel = case_when(
12     Vote == 1 ~ "Yes",
13     Vote == 2 ~ "No",
14     Vote == 3 ~ "Abstain",
15     Vote == 4 ~ "Present but did not vote",
16     Vote == 0 ~ "Absent",
17     TRUE ~ "Unknown"
18   ))
```

For this step, I needed to rename the column names for they are the same across the datasets. I also needed to rename the vote column to make sure they are clearly indicated as votes. In the next step I needed to pivot the data so that there is one row per MEP vote. Then I needed to convert the numeric codes for the votes into text.

- Create a summary table of counts of decision categories (e.g. Yes/No/Abstain/Present but did not vote/Absent) across all votes.

```
1 vote_summary <- rcv_ep1_long %>%
2   group_by(VoteLabel) %>%
3   summarise(Count = n()) %>%
```

```

4 arrange(desc(Count))
5 vote_summary

```

VoteLabel	Count
<chr>	<int>
1 Present but did not vote	109224
2 Not an MEP	103618
3 Absent	99753
4 Yes	88185
5 No	75171
6 Abstain	9577

4. Construct a new dataset that combines MEP-level information with their vote decisions from EP1 in long format (from part 3). Check for missingness.

```

1 mep_ep1 <- mep_list$EP1 %>%
2   rename(MEPID = 'MEP id') %>%
3   select(
4     MEPID,
5     'National Party',
6     'EP Group',
7     'NOM-D1',
8     'NOM-D2'
9   )
10 mep_ep1$MEPID <- as.character(mep_ep1$MEPID)
11 rcv_ep1_long$MEPID <- as.character(rcv_ep1_long$MEPID)
12 combined_ep1 <- left_join(
13   rcv_ep1_long,
14   mep_ep1,
15   by = "MEPID"
16 )
17 missing_summary <- combined_ep1 %>%
18   summarise(across(everything(), ~ sum(is.na(.))))
19 missing_summary

```

In the first step, I extracted EP1 metadata from the Excel sheets to keep only the relevant columns for each MEP. I ensured that the MEPID column had the same type (character) in both the metadata and the voting dataset so they could be merged correctly. I then combined the vote data with the metadata using a left join and checked the resulting dataset for missing values

```

A tibble: 1 × 12
MEPID MEPNAME    MS    NP    EPG VoteNumber  Vote VoteLabel
<int>   <int> <int> <int> <int>      <int> <int>    <int>
1      0      0      0      0      0          0      0        0
# 4 more variables: 'National Party' <int>, 'EP Group' <int>,

```

```
# 'NOM-D1' <int>, 'NOM-D2' <int>
```

5. Compute, for each EP group in EP1:

- The mean rate of Yes votes (Yes over Yes+No+Abstain) across all roll calls.
- The mean abstention rate.
- The mean vote preferences along the two contested dimensions (NOM-D1 and NOM-D2).

```
1 valid_votes <- combined_ep1 %>%
2   filter(VoteLabel %in% c("Yes", "No", "Abstain"))
3 ep_group_votes <- valid_votes %>%
4   group_by(EPG) %>%
5   summarise(
6     yes_rate = mean(VoteLabel == "Yes"),
7     abstention_rate = mean(VoteLabel == "Abstain"),
8     .groups = "drop"
9   )
10 ep_group_nom <- combined_ep1 %>%
11   filter(EPG != "0") %>%
12   select(EPG, MEPID, 'NOM-D1', 'NOM-D2') %>%
13   distinct() %>%
14   mutate(
15     'NOM-D1' = as.numeric('NOM-D1'),
16     'NOM-D2' = as.numeric('NOM-D2')
17   ) %>%
18   group_by(EPG) %>%
19   summarise(
20     mean_NOM_D1 = mean('NOM-D1', na.rm = TRUE),
21     mean_NOM_D2 = mean('NOM-D2', na.rm = TRUE),
22     .groups = "drop"
23   )
24 ep_group_summary <- left_join(ep_group_votes, ep_group_nom, by = "EPG")
25 ep_group_summary
```

In this step, I first filtered the combined EP1 dataset to include only valid votes, keeping rows where the VoteLabel was “Yes,” “No,” or “Abstain.” I then grouped these votes by European Parliament group (EPG) and calculated two measures of voting behavior for each group: the proportion of votes that were “Yes” and the proportion of abstentions. Separately, I extracted the NOMINATE scores for each MEP, removed any duplicate rows to ensure one row per MEP, converted the scores to numeric values, and computed the mean score on each dimension by EP group. Finally, I merged the vote statistics with the NOMINATE averages using a left join to produce a summary table showing both voting cohesion and ideological position for each EP group. showing both voting cohesion and ideological position for each EP group.

EPG	yes_rate	abstention_rate	mean_NOM_D1	mean_NOM_D2
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 C	0.415	0.0752	0.811	0.530
2 E	0.509	0.0215	0.512	-0.277
3 G	0.512	0.0697	0.280	-0.818
4 L	0.486	0.0632	0.409	-0.324
5 M	0.528	0.0800	-0.357	-0.201
6 N	0.581	0.0562	0.250	-0.386
7 R	0.457	0.265	-0.586	-0.0419
8 S	0.576	0.0574	-0.0980	0.261

>

Data Visualization

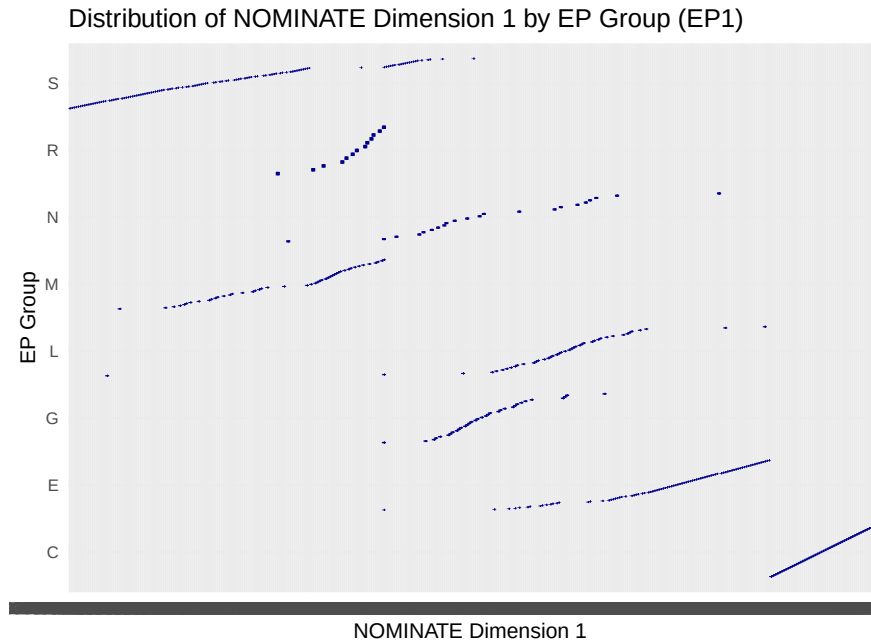
1. Plot the distribution of the first NOMINATE dimension by EP group, and explain any trends you see.

```

1 library(ggplot2)
2 epg_boxplot <- ggplot(
3   combined_ep1 %>% filter(!is.na('NOM-D1'), EPG != "0"),
4   aes(x = reorder(EPG, 'NOM-D1', FUN = median), y = 'NOM-D1')
5 ) +
6   geom_boxplot(fill = "skyblue", color = "darkblue") +
7   coord_flip() +
8   labs(
9     title = "Distribution of NOMINATE Dimension 1 by EP Group (EP1)",
10    x = "EP Group",
11    y = "NOMINATE Dimension 1"
12 ) +
13   theme_minimal(base_size = 14)
14 ggsave(
15   filename = "vote_graph_one.pdf",
16   plot = epg_boxplot,
17   width = 8,
18   height = 6,
19   units = "in"
20 )

```

This plot shows that voting behavior is associated with party affiliation. The same party groups tend to vote similarly.



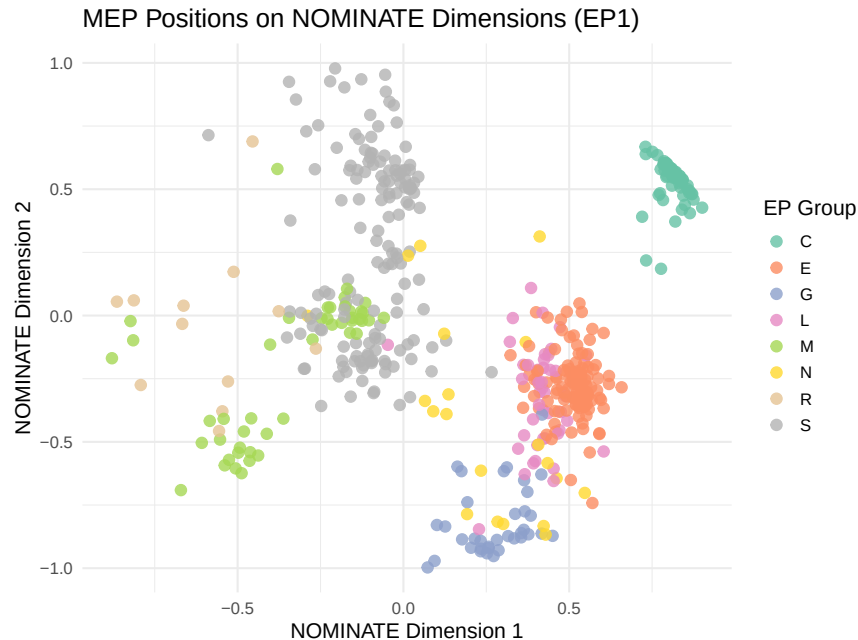
2. Make a scatterplot of *nomdim1* (x-axis) and *nomdim2* (y-axis), with one point per MEP and color by EP group.

```

1 library(ggplot2)
2 plot_data <- combined_ep1 %>%
3   filter(!is.na('NOM-D1'), !is.na('NOM-D2'), EPG != "0") %>%
4   distinct(MEPID, .keep_all = TRUE) %>%
5   mutate(
6     'NOM-D1' = as.numeric('NOM-D1'),
7     'NOM-D2' = as.numeric('NOM-D2')
8   )
9 nominate_scatter <- ggplot(plot_data, aes(x = 'NOM-D1', y = 'NOM-D2',
10   color = EPG)) +
11   geom_point(size = 3, alpha = 0.8) +
12   scale_color_brewer(palette = "Set2") +
13   labs(
14     title = "MEP Positions on NOMINATE Dimensions (EP1)",
15     x = "NOMINATE Dimension 1",
16     y = "NOMINATE Dimension 2",
17     color = "EP Group"
18   ) +
19   theme_minimal(base_size = 14) +
20   theme(legend.position = "right")
21 ggsave(
22   filename = "vote_graph_two.pdf",
23   plot = nominate_scatter,
24   width = 8,
25   height = 6,
26   units = "in"
27 )

```

In the scatterplot, MEPs from Group C and Group E cluster tightly, indicating that members of these groups are very aligned in their voting behavior. By contrast, Group S is more spread out across the NOMINATE dimensions, showing that its members are less cohesive and have more variation in their voting preferences.



3. Produce a boxplot of the proportion voting Yes by EP group to visualize cohesion.

```

1 library(ggplot2)
2 valid_votes <- combined_ep1 %>%
3   filter(VoteLabel %in% c("Yes", "No", "Abstain"))
4 mep_yes <- valid_votes %>%
5   group_by(MEPID, EPG) %>%
6   summarise(
7     prop_yes = mean(VoteLabel == "Yes"),
8     .groups = "drop"
9   )
10 vote_cohesion_plot <- ggplot(mep_yes, aes(x = reorder(EPG, prop_yes, FUN
11   = median), y = prop_yes, fill = EPG)) +
12   geom_boxplot() +
13   coord_flip() + # horizontal for readability
14   scale_fill_brewer(palette = "Set2") +
15   labs(
16     title = "Cohesion of EP Groups in EP1 (Proportion of Yes Votes)",
17     x = "EP Group",
18     y = "Proportion Voting Yes"
19   ) +
20   theme_minimal(base_size = 14) +
21   theme(legend.position = "none")
22 ggsave(

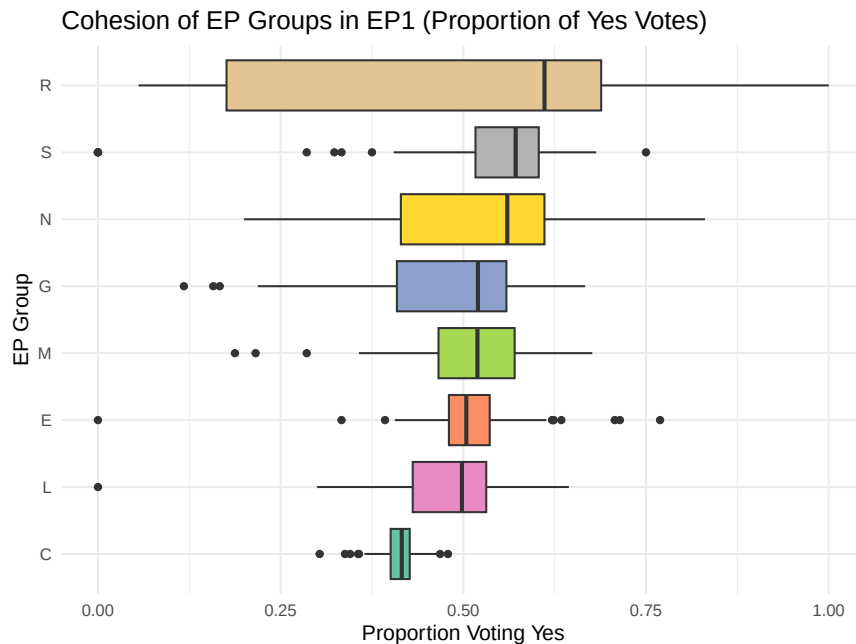
```

```

22 filename = "vote_graph_three.pdf",
23 plot = vote_cohesion_plot,
24 width = 8,
25 height = 6,
26 units = "in"
27 )

```

The boxplot shows substantial differences in voting cohesion across EP groups. EP Group R exhibits the widest spread in the proportion of Yes votes, indicating considerable internal disagreement among its members. In contrast, EP Group C has the narrowest spread, suggesting a high level of internal cohesion and more disciplined voting behavior.



4. Display the proportion voting *Yes* per year by national party using a bar plot.

```

1 library(dplyr)
2 party_yes <- combined_ep1 %>%
3   filter(VoteLabel %in% c("Yes", "No", "Abstain")) %>% # keep valid
4   group_by('National Party') %>%
5   summarise(
6     prop_yes = mean(VoteLabel == "Yes"),
7     .groups = "drop"
8   )
9 library(ggplot2)
10
11 ggplot(party_yes, aes(x = reorder('National Party', prop_yes), y = prop_
12   yes, fill = prop_yes)) +
13   geom_col() +

```

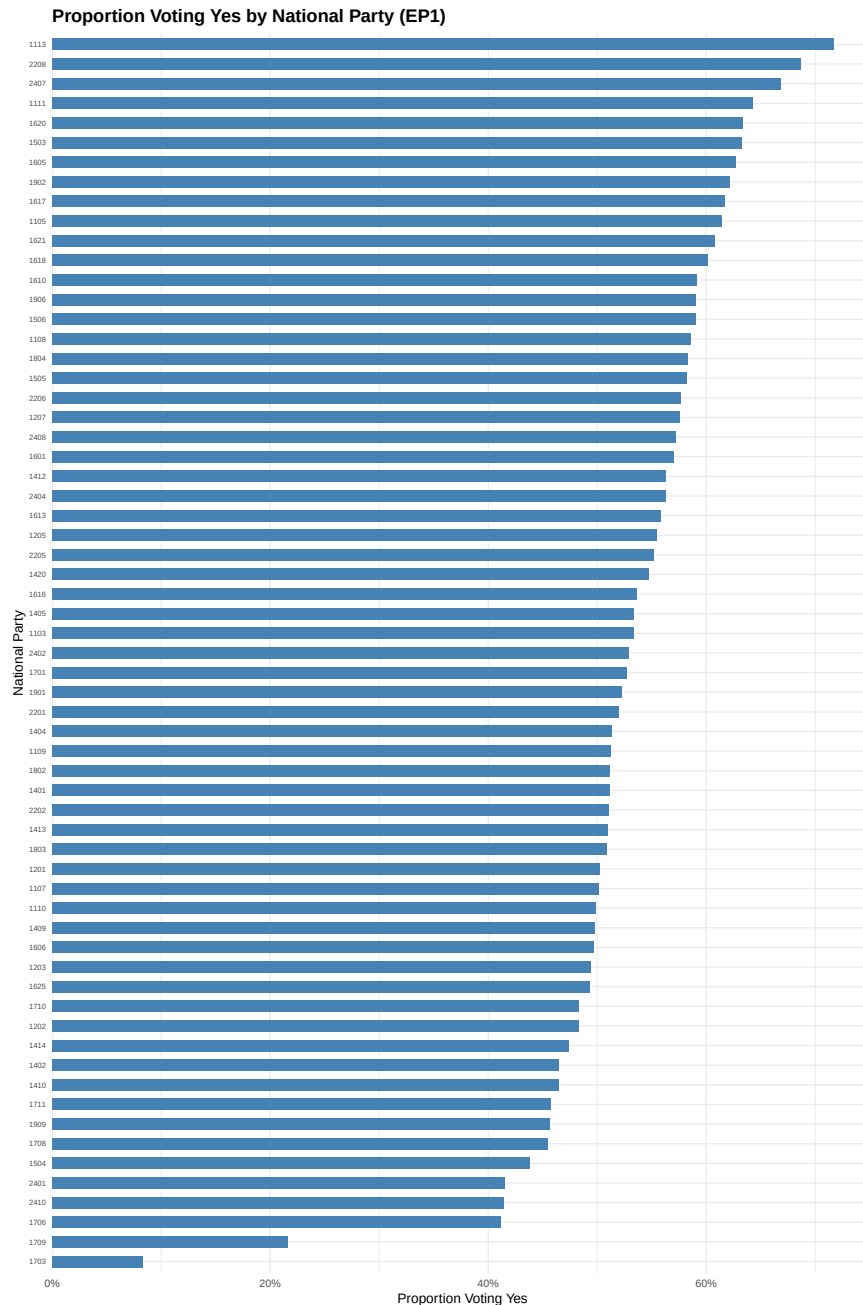


```

13 coord_flip() +
14 scale_fill_gradient(low = "lightblue", high = "darkblue") +
15 labs(
16   title = "Proportion of Yes Votes by National Party (EP1)",
17   x = "National Party",
18   y = "Proportion Voting Yes"
19 ) +
20 theme_minimal(base_size = 10) +
21 theme(
22   legend.position = "none",
23   axis.text.y = element_text(size = 3)
24 )
25
26 ggsave(
27   filename = "vote_graph_four.pdf",
28   width = 10,
29   height = 40,
30   units = "in"
31 )
32 #Graph 5
33 library(dplyr)
34 library(ggplot2)
35 ep_group_yes <- combined_ep1 %>%
36   filter(VoteLabel %in% c("Yes", "No", "Abstain"), EPG != "0") %>%

```

This plot is showing that most parties voted yes above 40 percent but there was two parties that has very low yes votes.



5. For each EP group, calculate the average *Yes* share per year and plot a line graph.

```

1   .groups = "drop"
2   )
3   p <- ggplot(ep_group_yes, aes(x = reorder(EPG, avg_yes), y = avg_yes,
4     fill = EPG)) +
5     geom_col() +
6     geom_text(aes(label = round(avg_yes, 2)), hjust = -0.1, size = 3) +
7     coord_flip() +
8     scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +

```

```

8   labs (
9     title = "EP Group Cohesion: Average Yes Votes (EP1)", # updated
10    title
11    x = "EP Group",
12    y = "Average Proportion Yes"
13  ) +
14  theme_minimal(base_size = 12) +
15  theme(legend.position = "none")
16  ggsave(
17    plot = p,
18    filename = "Vote_Graph_5.pdf",
19    width = 8, # width in inches
20    height = 6, # height in inches
21    units = "in"
22  )

```

The lowest yes votes were in the group C and the highest were in group N. These groups political affiliation was likely a reason behind the higher yes or no votes.

